

feature vector $x_{i,t}$

har_daily

$GBM: x \in \mathbb{R}^9 \text{ (own)} \cdot GBME: \mathbb{R}^{13} \text{ (own+E)} \cdot +graph: \mathbb{R}^{14} \text{ (own+E+g)}$

cross-firm graph $W \in \mathbb{R}^{N \times N}$
 $W = \text{top-}k(\text{corr}(\log pk)), \text{ row-norm (train)}$

har_weekly

har_monthly

rq (realized quarticity)	
mr_change	
mr_slope5	
mr_slope10	
mr_dev5	
mr_z22	
earn_prox	
earn_soon	
earn_pre	
earn_post	
$g_{i,t} = \sum_j W_{ij} pk_{j,t}$	

$\mathbb{R}^9$ own-history

$\mathbb{R}^4$ external (E)

$\mathbb{R}^1$ graph

Gamma-loss gradient
boosting (GBM)

$$F(x) = \sum_{m=1}^{300} \eta f_m(x)$$
$$f_m: \text{regression tree}$$

(31 leaves, $\eta = 0.05$)

loss: gamma deviance
= QLIKE + const
(3-seed ensemble)

$x_{i,t}$

$\hat{\sigma}_{i,t+h}^2$
Parkinson
variance forecast
 $h \in \{1, 5, 10, 22\}$

trained per market:
S&P 500 · HOSE

$$OHLCV_{i,t}$$

(stock i)

$$\rightarrow pk_{i,t}$$
$$= \frac{\ln(H/L)^2}{4 \ln 2}$$